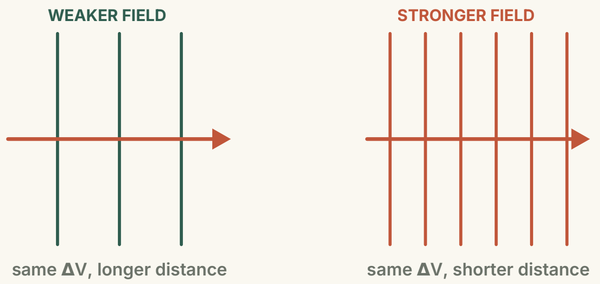
ELECTROSTATICS

Potential Gradient

How voltage changes across an electric field.

ESSENTIAL QUESTION

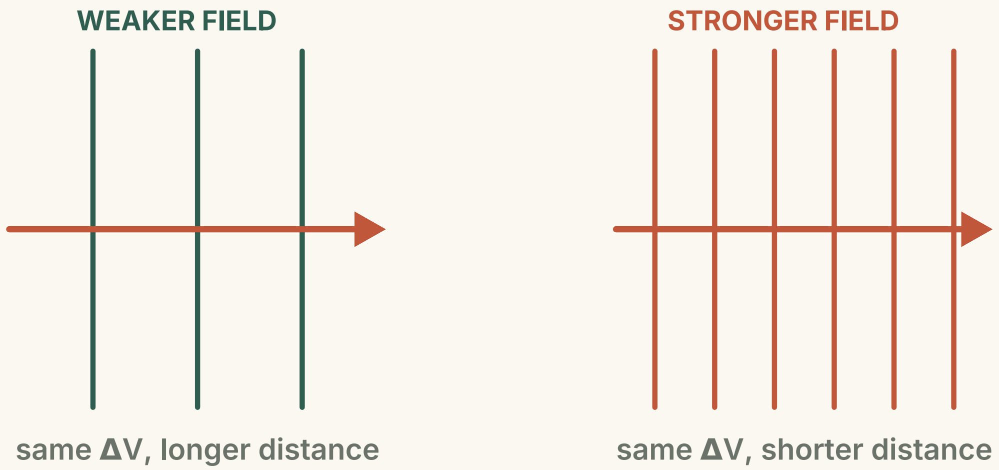
Why are equipotential lines closer together in a stronger field?



OBSERVE

One potential drop, two field strengths.

Equal-step equipotentials act like a visual ruler for the field.



DERIVE

Calculate the electric work two ways.

Uniform field; B is a distance d downfield from A.

FORCE × DISTANCE

$$W_{AB} = Fd = qEd$$

WORK PER UNIT CHARGE

$$U_{AB} = \frac{W_{AB}}{q} = Ed$$

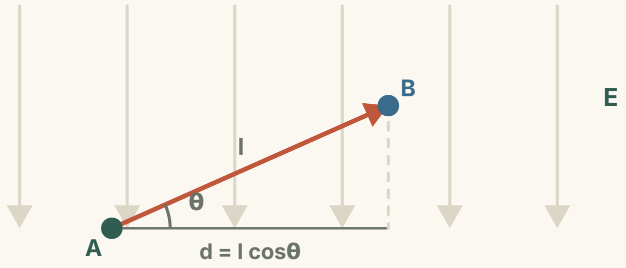
CONVENTION

$U_{AB} = V_A - V_B$, so U_{AB} is positive when B is downfield from A.

GENERALISE

The field only ‘sees’ the parallel component.

For an oblique displacement, project the journey onto the field direction.



PROJECT ONTO THE FIELD

$$d = l\cos\theta$$

THEN THE SAME RULE

$$U_{AB} = E\,l\cos\theta$$

Perpendicular motion contributes zero.

COMPARE

Direction controls the potential change.

Potential is a scalar, but displacement relative to E sets its change.

Move with E

V decreases — $\Delta V < 0$

Move across E

V unchanged — $\Delta V = 0$

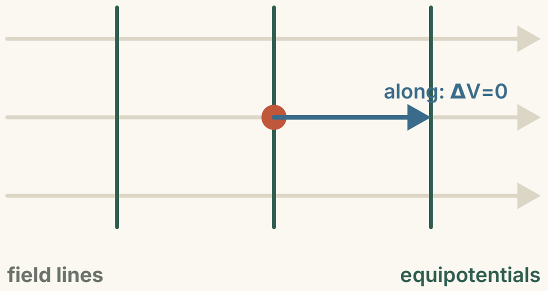
Move against E

V increases — $\Delta V > 0$

MAP

Equipotentials are perpendicular to the field.

Moving along one equipotential requires no electric work.



ALONG AN EQUIPOTENTIAL

$$\Delta V = 0 \;\Rightarrow\; W = q\Delta V = 0$$

The field points in the direction of the **fastest** potential decrease.

INTERPRET

Field strength is potential drop per unit distance.

This is the physical meaning of the potential gradient.

$$E = \frac{U}{d}$$

UNIT CHECK

$1\ \mathrm{V\ m^{-1}} = 1\ \mathrm{N\ C^{-1}}$, so the gradient is literally a field strength.

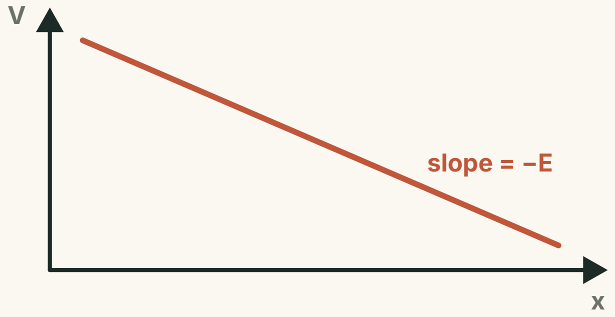
READ IT

A field of $200\ \mathrm{V\ m^{-1}}$ means potential falls 200 V for every metre along E.

GRAPH

A uniform field gives a constant gradient.

Potential decreases linearly with position along the field.



STEP BY STEP

$$\Delta V = -E\,\Delta x$$

IN THE LIMIT

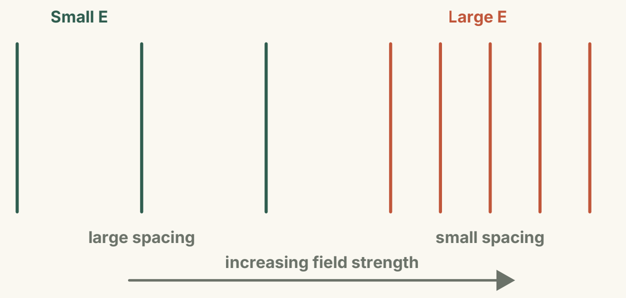
$$\frac{dV}{dx} = -E$$

The minus sign: potential decreases in the direction of E.

EXPLAIN

Why strong fields pack equipotentials tightly.

Compare neighbouring lines that differ by the same potential step.



FOR A FIXED ΔV

$$d = \frac{\Delta V}{E}$$

Small E → large spacing, potential changes slowly.

Large E → small spacing, potential changes fast.

CHECK

Three misconceptions to remove.

Read the equation as a relationship within a field, not as a cause.

“A larger U makes a larger E.”

No. E is a property of the existing field; U depends on the two points chosen.

“Any direction of falling V is E.”

No. E points toward the fastest decrease in potential.

“$U = Ed$ works for every path.”

No. The simple finite formula is exact only in a uniform field.

APPLY

Worked example: use the projection.

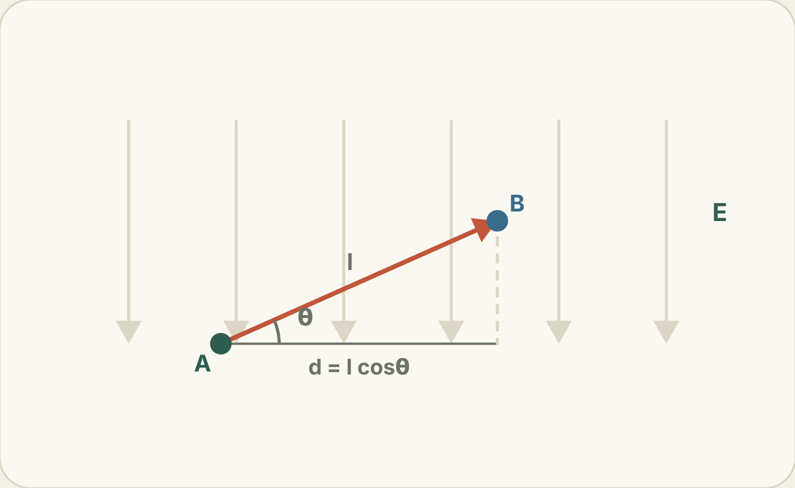
A uniform field has E = 250 N C⁻¹. Find U_AB for a 0.10 m step at 60° to the field.

1 **Project**

$$d = 0.10\cos 60^\circ = 0.050\ \mathrm{m}$$

2 **Calculate**

$$U_{AB} = Ed = 250 \times 0.050 = 12.5\ \mathrm{V}$$



TOOLKIT

Three ways to find a potential difference.

Choose the method that matches the information available.

FROM POTENTIALS

$$U_{AB} = V_A - V_B$$

When both point potentials are known.

FROM WORK

$$U_{AB} = \frac{W_{AB}}{q}$$

When electric work and charge are known.

FROM A UNIFORM FIELD

$$U_{AB} = Ed$$

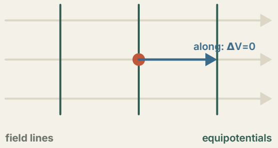
Use the projected downfield distance.

SUMMARY

Field strength is the potential gradient — the drop in V per unit distance.

AND IN A NON-UNIFORM FIELD

$$E = -\nabla V$$



In a uniform field U_AB = Ed, exact for the downfield projection.

Equipotentials crowd where the field is strong; they spread where it is weak.